OM SHAH

Security-focused Python Automation Engineer

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PROFILE

Cybersecurity graduate with three years of experience building production automation and security-focused software. Cut average case-processing time by 35% at Migrate Zone and independently built EightScope, a live external attack surface monitoring platform.

CORE SKILLS

Security External attack surface monitoring, vulnerability assessment, Nmap and NSE, Wireshark, DNS and TLS assessment, HTTP security headers, MITRE ATT&CK, OWASP Top 10

Engineering Python, REST APIs, PostgreSQL, JSON, CSV, OCR and PDF workflows, Excel automation, Git, HTML and CSS

Platforms Linux, Windows, VMware, AWS IAM and Security Groups, Burp Suite, Postman, Jira

EXPERIENCE

MIGRATE ZONE

Automation Engineer, Part-time — December 2024 - Present

Junior Automation Engineer, Part-time — July 2023 - November 2024

- Cut average case-processing time from approximately 8.0 hours to 5.3 hours by co-building Python workflows for document validation, file naming, risk flagging and case routing.
- Combined financial audit checks and Excel prefill into a single workflow, reducing duplicate reviews and repetitive data entry.
- Co-developed a PDF collation and OCR validation process that caught missing, expired and unreadable documents before lodgement.
- Built a lightweight monitor for government policy pages and wrote internal guides so non-technical staff could use the automation tools independently.

EIGHTSCOPE — 2026 - Present

Independent Developer

eightscope.au

- Built and launched a live platform that maps a domain's public attack surface, identifies externally visible security risks and tracks changes between scans.
- Implemented DNS ownership verification, subdomain discovery, hostname-level asset inventories and repeatable scan workflows.
- Added safeguards that prevent scans from targeting private, loopback, link-local and reserved addresses.
- Produced structured findings with severity, confidence, technical evidence and remediation guidance.
- Added scan coverage across DNS, email authentication, TLS, HTTP security headers, exposed services, public files, detected technologies and possible CVE signals.
- Created PostgreSQL-backed scan history and asset-aware comparison between completed scans.

ITONKEY — August 2024 - October 2024

Python Backend Developer Intern, Part-time

- Developed a Python system that turned structured business data into branded, multi-page PDF invoices.
- Contributed stock-tracking and reporting features to an inventory-management platform for small businesses.

SELECTED PROJECTS

SENTINEL — UTS Capstone, 2025

Automated Network Vulnerability Scanner

- Created a Python vulnerability scanner using Nmap and NSE to discover hosts, identify services and surface possible CVE matches.
- Improved fingerprinting across SMB, RDP, nested Nmap output and host-level script results.
- Generated risk-prioritised JSON findings grouped by asset, including commands for manual verification.

DNS TUNNELLING DETECTION LAB — Independent Security Lab, 2025

- Created an isolated VMware lab to study DNS tunnelling and capture network evidence using Wireshark.
- Produced defensive detection logic based on query entropy, subdomain length, burst timing and repeated-domain behaviour.
- Mapped the behaviour to MITRE ATT&CK T1048.003 and presented the captured PCAP through an interactive browser walkthrough.

EDUCATION

BACHELOR OF CYBERSECURITY — 2023 - 2026

University of Technology Sydney

Relevant study: Network Security, System Security, Cloud Security, Digital Forensics, Incident Response, Cyber Threat Intelligence, Cryptography and Information Security Management